

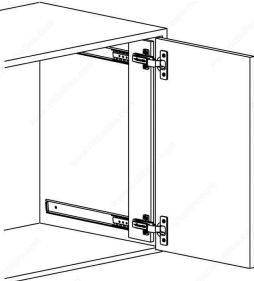
My Design: Water Filter
Armin Zarbaft, Jared Wagner, John Estock

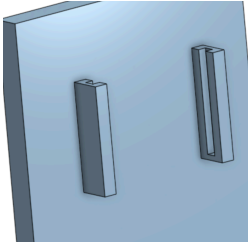
Element D: Design Concept Generation, Analysis, and the Selection

Design theory:

After completing research and discussing with stakeholders and other groups, we directed the focus of our project on organizing the sensory space (utilizing the middle island and cubbies) and introducing tactile panels.

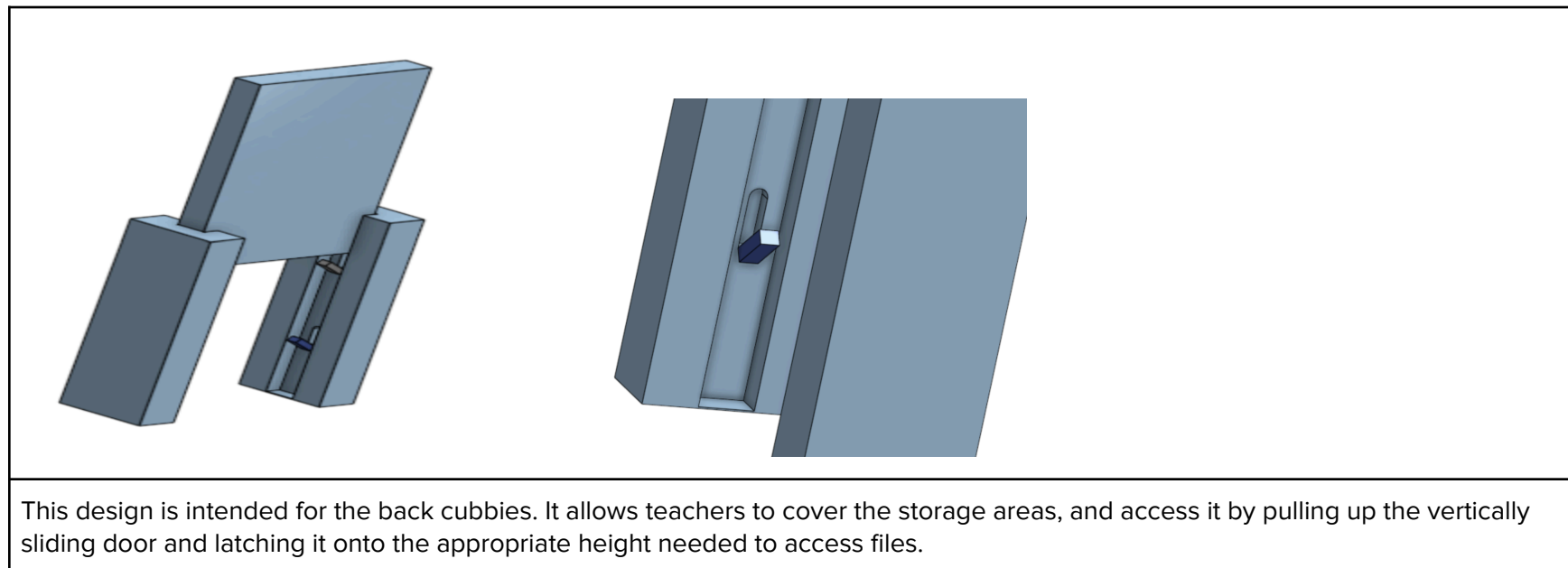
Morphological chart:

Functions/ Designs	Design 1	Design 2	Design 3	Design 4
Middle island door and cubby covers	Sliding doors 	Doors on hinges 	Pocket doors 	Vertical wooden rails 

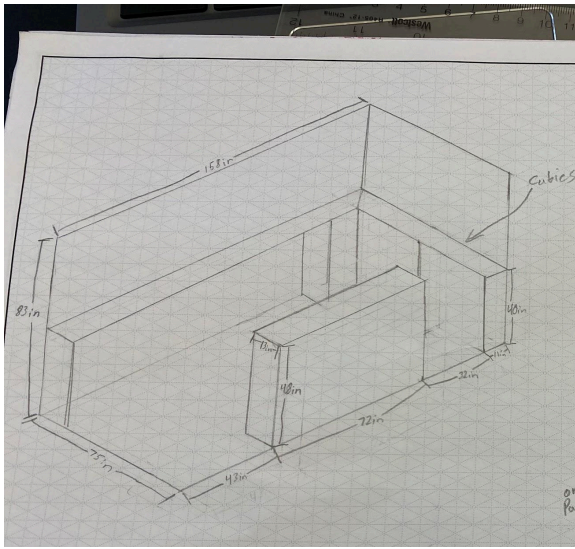
Functions/ Designs	Design 1	Design 2	Design 3	Design 4
Appealing tactiles(measured by kids interest)				
Interchangeability of tactiles and future projects	Wooden Slot 	Chain cut out 	French Cleat 	Screwed in 
Accessibility (locking)	Deadbolt 	Traditional door lock unlocked with key	Electronic keyless lock	

Functions/ Designs	Design 1	Design 2	Design 3	Design 4
				

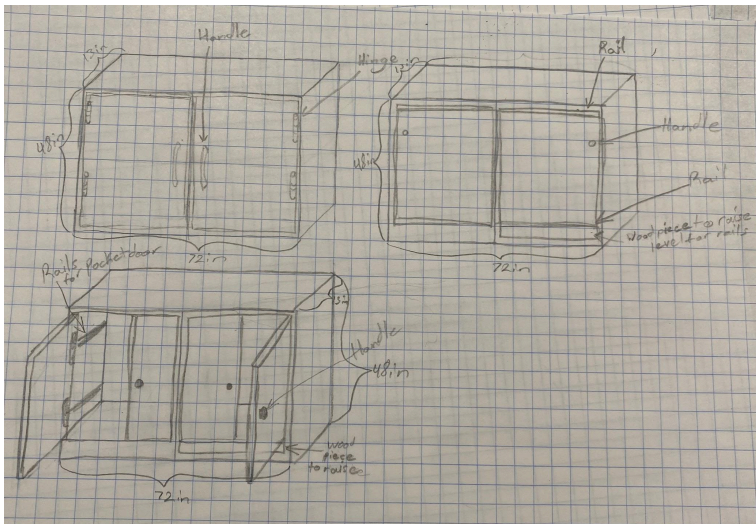
Vertical wooden rails for cubby door:



Sketch of room-

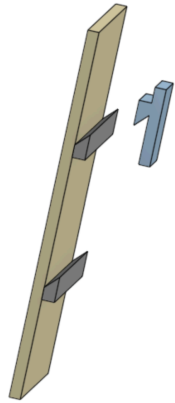


Sketches of designs for middle island doors-

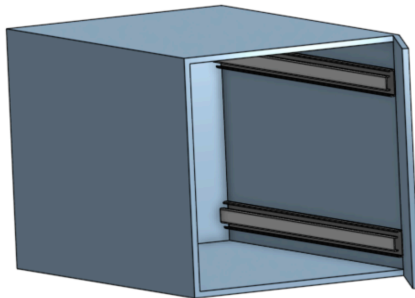


Conceptual Design 1:

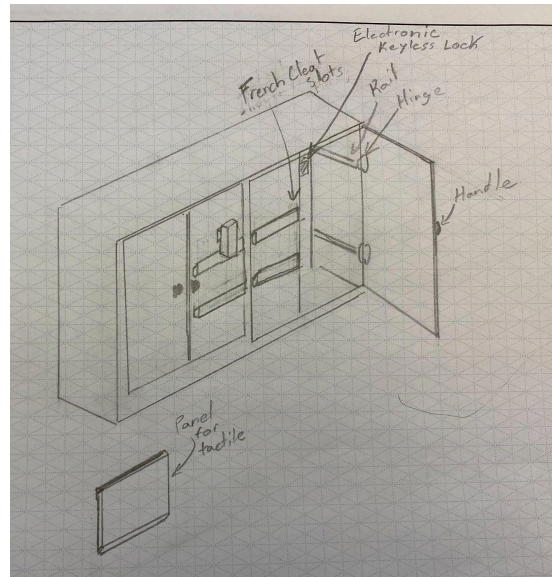
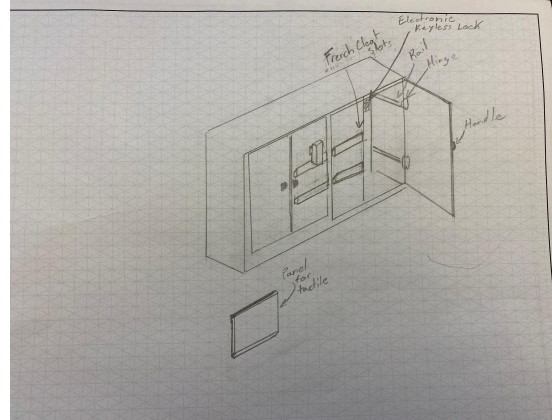
Panel:



Door:



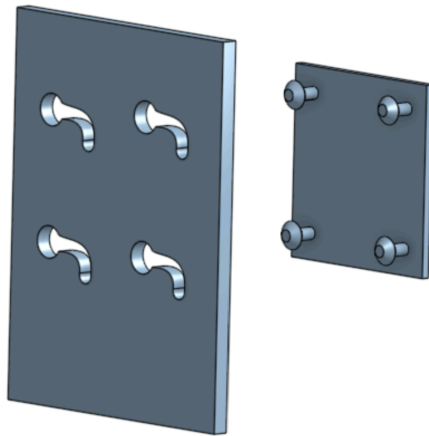
Sketches



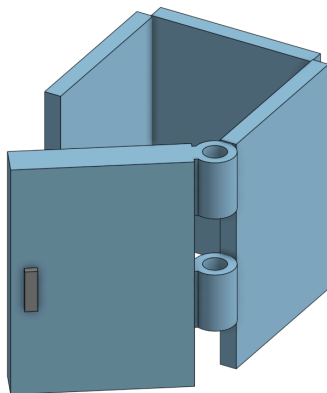
This conceptual design uses a pocket door for the middle island and a french cleat locking system for the interchangeable tactile panels. The pocket door allows for someone to swing it open like a typical cabinet door, but with the extra benefit of being able to be slid in along a rail so that it is pocketed within the cabinet/middle island. This is useful as it not only provides cover for whatever teachers have stored in the middle island, but also reduces the problem of the sensory area being cramped. Having it so that the door can be stored at will removes clutter and the potential hazards that come with it. The main tradeoff of the pocket door is that it is more complicated than some of the other designs, requiring more materials and more money. It also runs the risk of damaging the tactiles attached to the door when the door is being slid on its rail back into the cabinet. To minimize this issue, we paired the door with the french cleat for the attachment of tactiles. The french cleat keeps the tactiles pretty flat against the surface of the door, minimizing collisions from the tactile to the inner wall of the middle island. This design also has the issue that kids can more easily remove the tactiles.

Conceptual Design 2:

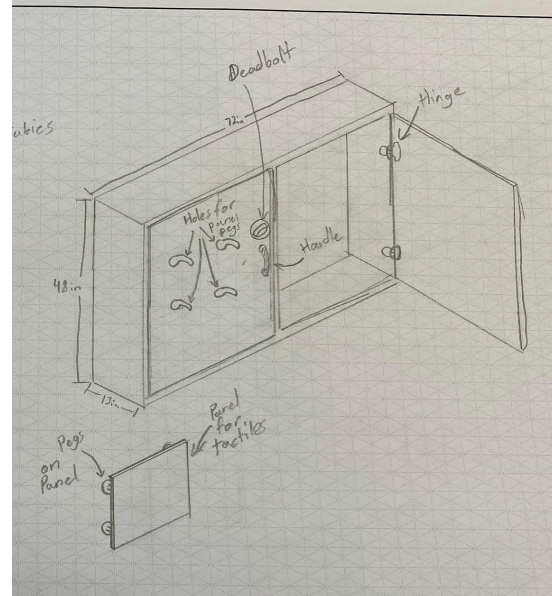
Panel:



Door:



Sketches

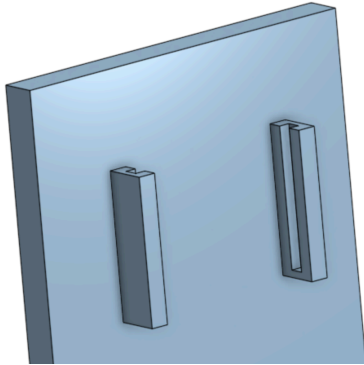


This conceptual design uses hinged doors for the middle island and a 'ball and chain cutout' locking system for the interchangeable tactile panels. The door is swung open like a typical cabinet door, allowing for items to be stored in the middle island and covered. Because this design is like typical cabinet doors, it is easy and simple to use. The 'ball and chain' also allows for tactiles to be locked and unlocked into place at will, promoting interchangeability while keeping kids from removing the tactiles when they aren't supposed to. One drawback of the hinged doors is that when swung open, the doors add to the clutter in the sensory space and are a hazard for people, especially children, who are in the area. Because the island is close to the ground, the doors could be a tripping hazard.

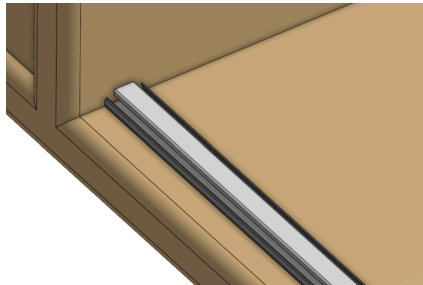
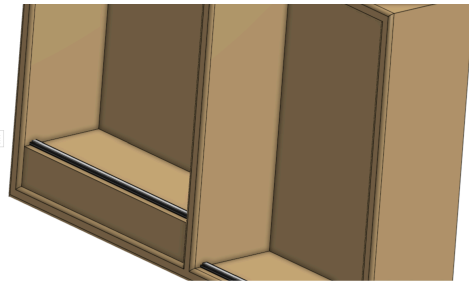
The 'ball and chain' tactile has a tradeoff in its requirements. In order to lock the tactiles, holes must be drilled into the newly added middle island doors, removing the potential for a change in usage of the doors in the future.

Conceptual Design 3: sliding door

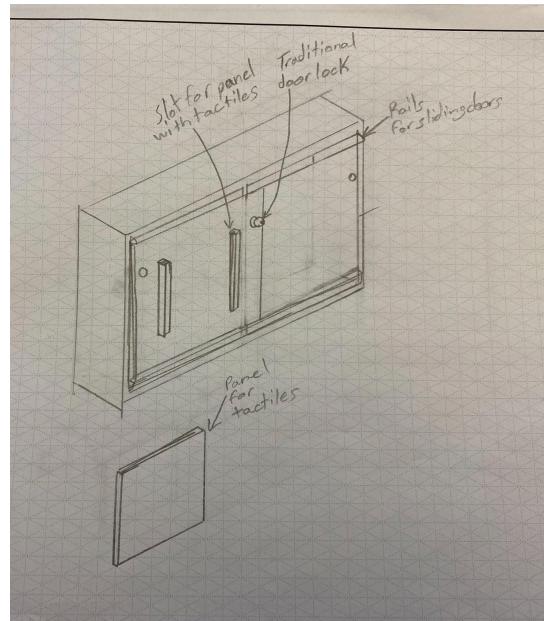
Panel:



Door:



Sketches



This conceptual design features a door that opens and closes by sliding on a horizontal rail and a vertical rail for interchangeable tactile panels. The door having the ability to slide reduces clutter brought by typical hinged doors while keeping items stored. The vertical rails used for the tactiles ensures interchangeability, but is restrictive in its requirement for the panels to be a specific thickness.

The horizontal rails for the door are not the most durable, and also are costly.

Decision Matrix:

Note: Grayed out qualities of design are uniform across our three conceptual designs. Because we are handling multiple projects in one and the conceptual designs do not include all aspects of the project that we must design, the qualities of design do not match perfectly with conceptual designs.

QODs	Weight	Design 1		Design 2		Design 3	
		Score	Weighted	Score	Weighted	Score	Weighted
More storage space and improved organization	20%	5	1.00	4	0.80	5	1.00
Appealing tactiles	20%	5	1.00	5	1.00	5	1.00
Interchangeable	15%	4	0.60	3	0.45	3	0.45
Accessibility (locking)	15%	5	0.75	5	0.75	5	0.75
Cost	10%	2	0.20	4	0.40	3	0.30
Durability	10%	4	0.40	3	0.30	2	0.20
Uniform design	10%	5	0.50	4	0.40	3	0.30
Total	100%		4.45		4.10		4.00

Decision Rationale:

We felt that the most important aspects of our various designs included improving the organization and storage efficiency of the sensory space and creating appealing tactiles for the kids to interact with. Accordingly, we weighted both qualities 20%. After that, we prioritized interchangeability of the tactiles at 15% as we felt it was very important for whatever changes and additions we make to be adjusted, moved, and stored over time. Also at 15% we had accessibility, which measures a design's ability to allow teachers to access stored items while keeping kids from those same items. Next was cost, durability, and uniform design weighted at 10% each as they can be categorized as the 'wants' of our project rather than the 'needs'.

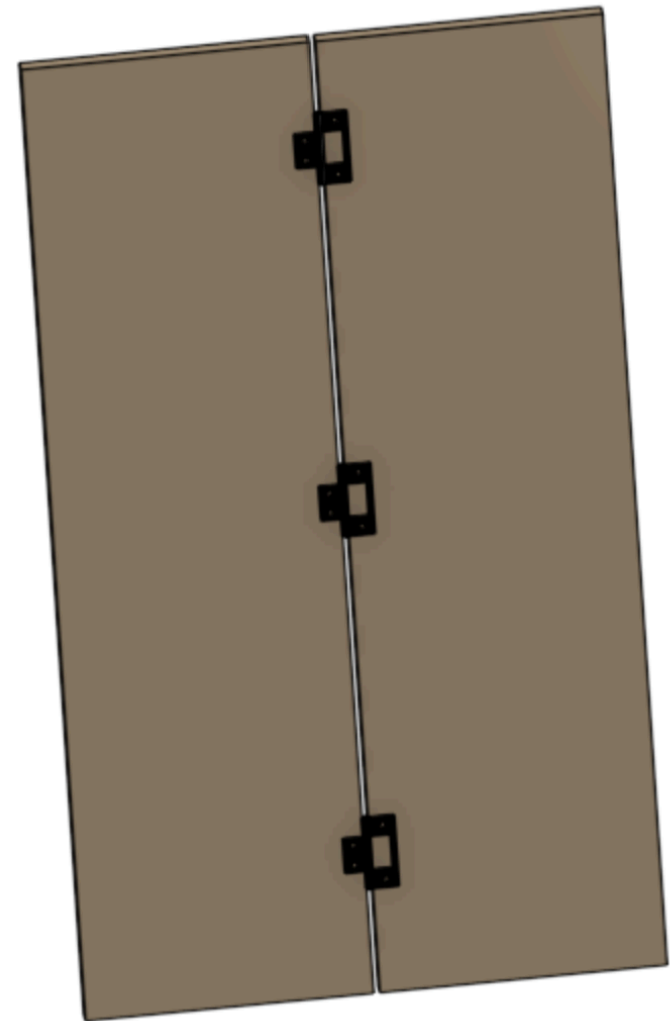
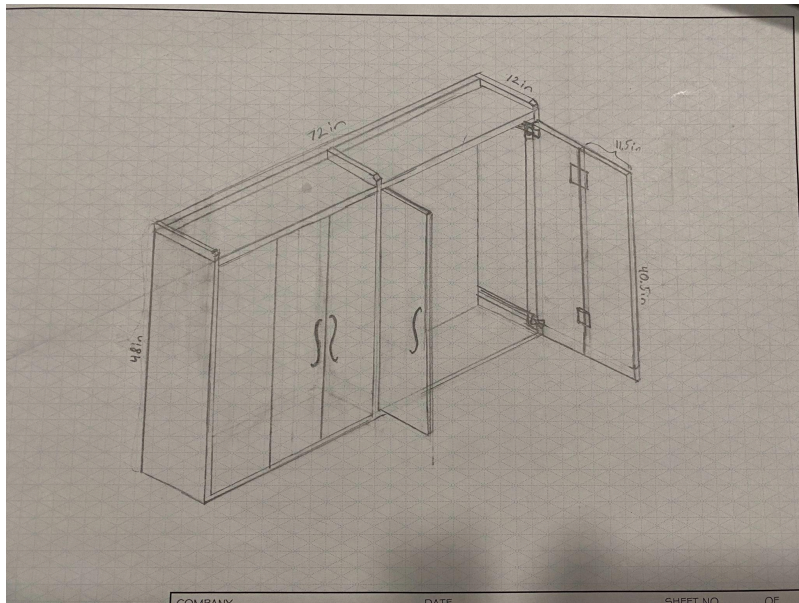
After scoring each design in the weighted categories, design 1 scored highest at 4.45 compared to design 2's score of 4.10 and design 3's score of 4.00. Because of this scoring, we are leaning towards moving forward with design 1 which consisted of a french cleat tactile on a pocket island door. Design 1 scores well in every category except cost, making it a very viable option if we can prepare proper funds in enough time for what we need. Designs 2 and 3 suffered in scoring largely due to their lack of interchangeability. Comparatively, design 1 scored well in interchangeability, largely due to the attributes of the french cleat.

The following is an edit made after making the initial design decision:

After talking with stakeholder Ms. Brennan on January 26th, 2024, who is the 'head' teacher of the special needs classroom, we have decided to go with the 'ball and chain' tactile panels from design 2 and the pocket door design from design 1. Ms. Brennan was very enthusiastic about having pocket doors, as this would remove the hazard of an opened door in the area. She emphasized that having that open door in the walk space was definitely a concern, and thus the pocket door would be optimal. Similarly, Ms. Brennan emphasized that she was in favor of the "ball and chain" as it seemed to have the best locking system, especially compared to that of the french cleat. Ms. Brennan also approved of the panel and door designs in design 3, but definitely showed preference for the 'ball and chain' and pocket door.

Concerns after talking with stakeholders:

Upon visiting the room again, we noted that the island and cubbies were pretty shallow, making the original pocket door design impossible without alteration. Because of this, we had to create a new sketch for a design, as well as a CAD adaptation allowing the pocket door to fold onto itself and still be stored within the storage space. Those sketches/CAD can be seen below. Also seen below is the rail we are getting for our pocket door.





During the meeting with Ms. Brennan, it was also clarified that we cannot touch the left cubby on the back wall, and it would be her preference if we do not touch the right cubby on the back wall. The two cubbies she was talking about are shown in the picture below.



This image shows the two back cubbies that have restrictions with covering

This would cut costs as we wouldn't have to produce as many door covers, but we now need to make sure that we have enough storage space in the cubbies and island we will be covering in order to clear up room needed for other sensory projects from other

groups. For these covers, we are leaning towards a salmon ladder design, which was described and shown under the morphological chart.

The two cubbies we will be working with are pictured below.



This image shows the two side cubbies that we need to provide covers for

After talking to Ms. Brennan, we have decided to complete our project in two separate phases. Phase 1 would consist of designing the pocket door for the middle island. Phase 2 would then consist of our salmon ladder cover for the side panels. Phase 3, which as noted under the manufacturing plan below, may not fit the timeline for this project. This phase consists of creating and implementing tactile panels on the middle island.

Splitting the project into phases should ensure that we do not leave anything unfinished should we face any unforeseen problems or delays. By prioritizing completion of phase 1, we are implementing an outline for optimizing space by combining storage covers with sensory panels. The interchangeability in our project should in the future allow for groups to make additions or adjustments to our system of panels that correspond with the changing demands of stakeholders.

Additionally, we are not focusing on locking mechanisms for the cabinets as we deemed them low priority.

Manufacturing Plan:

February 2024



Month ▾







	MON 29	TUE 30	WED 31 Manufacturing plan due ● 1:30pm Meeting with Ms. Brennan	THU Feb 1 Home Depot Visit and Research	FRI 2 Bill of materials	SAT 3
	5 Element D (OnShape drawing of new c ● 1:30pm Element E	6	7 Element F	8	9	10 Gather materials for construction
	12 Construction of prototype starts	13	14	15	16	17
	19	20	21	22	23 Middle Island Wood Work Done	24
	26	27	28	29	Mar 1 Hinges, rails, etc on prototype	2

March 2024

🔍 ? ⚙️ Month ▼ 📅

	MON 26	TUE 27	WED 28	THU 29	FRI Mar 1
					Hinges, rails, etc on prototype
	4	5	6	7	8
	11 Testing/adjustments finished	12 Initiate implementation of design to isl	13	14	15

Phase 2, which consisted of providing covers for the sensory room cubbies, and phase 3, which consisted of implementing tactile panels into the middle island covers, were not included in the original manufacturing plan. The reason the project was split into phases was because of innate uncertainties we have with the timeline of the project. This is part of meeting the stakeholder's need that all started projects concerning the infrastructure of the sensory room be finished.